

Stability Analysis of Networked Predictive Control Systems with Random Network Delay

Yuanqing Xia, J. Chen, G. P. Liu and D. Rees

Abstract—The paper is concerned with the design and stability analysis of networked control systems with random network delay, which are described in a state-space form. It presents a new control scheme, which is termed networked predictive control. This scheme mainly consists of the control prediction generator and network delay compensator. The control prediction generator provides a set of future control predictions to make the closed-loop system achieve the desired control performance and the network delay compensator removes the effects of the network transmission delay. The stability criteria of the closed-loop networked predictive control systems are analytically derived for fixed. In case of random network delay, a switched system's method has been proposed, which solves the stochastic problem in a deterministic way.

Index Terms—Networked control system, predictive control, random network delay, stability.

I. INTRODUCTION

In the last decade, network technology has dramatically been developed. Recently, more and more network technology has been applied to control systems [1]. Now, networked control is a new area in control systems. Particularly, Internet based control systems allow remote monitoring and adjustment of plants over the Internet. This makes the control systems benefit from the ways of retrieving data and reacting to plant fluctuations from anywhere around the world at any time. The networked control has also opened up a complete new range of real-world applications, namely tele-manufacturing, tele-surgery, museum guidance, traffic control, space exploration, disaster rescue, and health care.

In networked control systems (NCS), the plant, controller, sensor, actuator and reference command are connected through a network. As the structure of networked control systems is different from that of tradition control systems, there exist various specific problems in networked control systems, for example, network delay, loss of data packets, network security and safety [2].

As there are more and more applications of networked control systems in industry, such as traffic, communication, aviation and spaceflight, more attention in this area has been paid to the design and analysis of NCSs [3]. The stability problem of closed-loop NCS in the presence of network delays and data packet drops have been addressed in [4]. To

reduce network traffic load, a sampled-data NCS scheme has been presented and some necessary and sufficient conditions for global exponential stability of the closed-loop systems via state/output feedback, without/with network delays have been established in [5]. Some issues related to network bandwidth constraints and network traffic congestion in NCSs have also been studied in [6], [7] and [8]. Internet based control has also been applied to practical systems, for example, Internet-based process control [2], Internet based control systems as a control device [9], Internet robots [10], Internet based multimedia education [11], process monitoring and optimization via the web [12].

Although much research work has been done in networked control systems, most work has ignored a very important feature of networked control systems. This feature is that the communication networks can transmit a packet of data at the same time, which is not done in traditional control systems. In [13], the predictive control for networked control systems has been proposed to deal with networked systems in polynomial form. However, the predictive control for networked control systems has not been considered, which is more general. This paper makes full use of this network feature and proposes a new networked control scheme - networked predictive control, which can overcome the effects caused by network delay.

The paper is organized as follows. Section II presents a networked predictive control scheme such that the closed-loop system is asymptotically stable. Section III details the stability analysis of closed-loop networked predictive control systems. Some conclusion remarks are given in Section IV.

II. NETWORKED PREDICTIVE CONTROL FOR SYSTEMS WITH NETWORK DELAY

To overcome unknown network transmission delay, a networked predictive control scheme is proposed. It mainly consists of a control prediction generator and a network delay compensator. The control prediction generator is designed to generate a set of future control predictions. The network delay compensator is used to compensate the unknown random network delay. This paper considers the case where the system's sensor is near to the controller. So, the transmission delay in the feedback channel is not discussed.

Consider a MIMO discrete system described in the following state space form

$$\begin{aligned}x_{t+1} &= Ax_t + Bu_t \\ y_t &= Cx_t\end{aligned}\tag{1}$$

Yuanqing Xia and J. Chen is with the Department of Automatic Control, Beijing Institute of Technology, Beijing 100081, China. Email: xia_yuanqing@163.net; chenjie@bit.edu.cn. The work of Yuanqing Xia was supported by the National Natural Science Foundation of China under Grant 60504020 and Excellent young scholars Research Fund of Beijing Institute of Technology 2006y0103.

Yuanqing Xia, G. P. Liu, D. Rees and W. Hu are with the School of Electronics, University of Glamorgan, Pontypridd, CF37 1DL, UK. Email: xia_yuanqing@glam.ac.uk; gpliu@glam.ac.uk; drees@glam.ac.uk.

where $x_t \in R^n$, $u_t \in R^m$, and $y_t \in R^l$ are the system state, input, and output vectors, respectively, A , B , C are matrices of appropriate dimensions. For the simplicity of stability analysis, it is assumed that the reference input of the system is zero and the following assumptions are made.

Assumption 1: The pair, (A, B) , is completely controllable, and the pair, (A, C) is completely observable.

Assumption 2: The number of consecutive data dropouts must be less than N_1 (an positive integer).

Assumption 3: The upper bound of the network delay is not greater than N (an positive integer).

Remark 1: In a real NCS, if the data packet does not arrive at a destination in a certain transmission time, it means this data packet is lost, based on the commonly used network protocols. From the physical point of view, it is natural to assume that only a finite number of consecutive data dropouts can be tolerated in order to avoid that the NCS becomes open-loop. Thus, the number of consecutive data dropouts should be less than a finite number N_1 . Similarly, the network delay should also be bounded by a finite number N . Clearly, the upper bound number of consecutive data dropouts should not be greater than the upper bound of the network delay (*i.e.* $N_1 \leq N$).

The state observer is designed as

$$\hat{x}_{t+1|t} = A\hat{x}_{t|t-1} + Bu_{t|t} + L(y_t - C\hat{x}_{t|t-1}) \quad (2)$$

where $\hat{x}_{t+1|t} \in R^n$ and $u_{t|t} \in R^m$ are the one-step ahead state prediction and the input of the observer at time t and matrix $L \in R^{n \times l}$, which can be designed using observer design approaches. The estimator of the state

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + M(y_t - C\hat{x}_{t|t-1}) \quad (3)$$

where $\hat{x}_{t|t-1} \in R^n$ is the state observer. The state predictions at time t are constructed as

$$\begin{aligned} \hat{x}_{t+1|t} &= A\hat{x}_{t|t-1} + Bu_{t|t} + L(y_t - C\hat{x}_{t|t-1}) \\ \hat{x}_{t+2|t} &= A\hat{x}_{t+1|t} + Bu_{t+1|t} \\ &\vdots \\ \hat{x}_{t+N|t} &= A\hat{x}_{t+N-1|t} + Bu_{t+N-1|t} \end{aligned} \quad (4)$$

Assume that the controller is of the following form:

$$u_{t|t} = K\hat{x}_{t|t} \quad (5)$$

where $K \in R^{m \times n}$ is the the state feedback control matrix to be determined using modern control theory. Then, the control predictions are generated by

$$u_{t+k|t} = K\hat{x}_{t+k|t}, \text{ for } k = 0, 1, 2, \dots, N \quad (6)$$

Thus, it follows from equation (4) that

$$\hat{x}_{t+k|t} = (A + BK)^{k-1} \hat{x}_{t+1|t} \quad (7)$$

Based on equations (2), (3) and (6), it can be shown that

$$\begin{aligned} \hat{x}_{t+k|t} &= (A + BK)^{k-1} (BKMC + LC)x_t + (A + BK)^{k-1} \\ &\quad \times (A + BK - LC - BKMC)\hat{x}_{t|t-1} \end{aligned} \quad (8)$$

and

$$\begin{aligned} u_{t+k|t} &= K\hat{x}_{t+k|t} \\ &= K(A + BK)^{k-1} (BKMC + LC)x_t + K(A + BK)^{k-1} (A + BK - LC - BKMC)\hat{x}_{t|t-1} \end{aligned} \quad (9)$$

In order to compensate the network transmission delay, a network delay compensator is proposed. A very important characteristic of the network is that it can transmit a set of data at the same time. Thus, it is assumed that predictive control sequence at time t is packed and sent to the plant side through a network. The network delay compensator chooses the latest control value from the control prediction sequences available on the plant side. For example, if the following predictive control sequences are received on the plant side:

$$\begin{aligned} &[u_{t-k_1|t-k_1}^T, u_{t-k_1+1|t-k_1}^T, \dots, u_{t|t-k_1}^T, \dots, u_{t+N-k_1|t-k_1}^T]^T \\ &[u_{t-k_2|t-k_2}^T, u_{t-k_2+1|t-k_2}^T, \dots, u_{t|t-k_2}^T, \dots, u_{t+N-k_2|t-k_2}^T]^T \\ &\vdots \\ &[u_{t-k_t|t-k_t}^T, u_{t-k_t+1|t-k_t}^T, \dots, u_{t|t-k_t}^T, \dots, u_{t+N-k_t|t-k_t}^T]^T \end{aligned} \quad (10)$$

where the control values $u_{t|t-k_i}$ for $i = 1, 2, \dots, t$, are available to be chosen as the control input of the plant at time t , the output of the network delay compensator will be

$$u_t = u_{t|t-\min\{k_1, k_2, \dots, k_t\}} \quad (11)$$

In fact, using the networked predictive control scheme presented in this section, the control performance of the closed-loop system with network delay is very similar to that of the closed-loop system without network delay.

III. STABILITY ANALYSIS OF NETWORKED PREDICTIVE CONTROL SYSTEMS

Similar to traditional control systems, closed-loop stability is also the most important issue in the design of networked control systems. This section considers the stability of networked control systems for two cases: one is the the case of the fixed network transmission delay and the other is the case of the random network transmission delay.

It is assumed that the network delay i is constant. The network delay compensator is in the form of

$$u_t = u_{t|t-i} \quad (12)$$

In particular, the augmented system without time-delay, *i.e.*, $u_t = u_{t|t}$, can be described as follows:

$$\begin{aligned} \hat{x}_{t+1|t} &= (A + BK - LC - BKMC)\hat{x}_{t|t-1} \\ &\quad + (BKMC + LC)x_t \\ x_{t+1} &= (A + BKMC)x_t + (BK - BKMC)\hat{x}_{t|t-1} \end{aligned} \quad (13)$$

Proposition 1: The poles of the augmented system (13) can be assigned by state feedback and observer design, separately.

Proof: From (13), the augmented system can be written as

$$\begin{bmatrix} x_{t+1} \\ \hat{x}_{t+1|t} \end{bmatrix} = \Psi \begin{bmatrix} x_t \\ \hat{x}_{t|t-1} \end{bmatrix} \quad (14)$$

where

$$\Psi = \begin{bmatrix} A + BKMC & BK - BKMC \\ BKMC + LC & A + BK - LC - BKMC \end{bmatrix}.$$

Applying the state transformation,

$$\begin{bmatrix} x'_{t+1} \\ \hat{x}'_{t+1|t} \end{bmatrix} = \begin{bmatrix} I & -I \\ 0 & I \end{bmatrix} \begin{bmatrix} x_{t+1} \\ \hat{x}_{t+1|t} \end{bmatrix} \quad (15)$$

gives

$$\begin{bmatrix} x'_{t+1} \\ \hat{x}'_{t+1|t} \end{bmatrix} = \begin{bmatrix} A - LC & 0 \\ BKMC + LC & A + BK \end{bmatrix} \begin{bmatrix} x'_t \\ \hat{x}'_{t|t-1} \end{bmatrix} \quad (16)$$

It can be concluded that the poles of the augmented system are determined by $A + BK$ and $A - LC$, separately. Under Assumption 1, the poles of the augmented system can be arbitrarily assigned. ■

It is clear from Proposition 1 that system (13) is the normal observer based state-feedback control system. Let $M_1 = BK(A + BK)^{i-1}(BKMC + LC)$, $M_2 = BK(A + BK)^{i-1}(A + BK - LC - BKMC)$, $M_3 = BKMC + LC$, $M_4 = A + BK - LC - BKMC$, $M_5 = BK(A + BK)^{i-1}$, $M_6 = A - BKMC - LC$, then the first main result in this paper is presented as follows:

Theorem 1: For the networked predictive control systems (1) with constant network delay $i \in Z^+$ (the set of positive integers), the closed-loop system is stable if and only if all eigenvalues of the following matrix are within the unit circle.

$$\begin{bmatrix} 0 & 0 & \cdots & 0 & 0 & M_5 \\ I & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & I & A & 0 \\ 0 & 0 & \cdots & 0 & M_3 & M_4 \end{bmatrix} \quad (17)$$

or

$$\begin{bmatrix} 0 & 0 & \cdots & 0 & 0 & M_5 \\ I & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & I & M_6 & -BK \\ 0 & 0 & \cdots & 0 & BKMC + LC & A + BK \end{bmatrix} \quad (18)$$

where the above two matrices belong to $R^{(i+1)n \times (i+1)n}$

Proof: From (9) and (12), the closed-loop system can be written as

$$\begin{aligned} x_{t+1} &= Ax_t + Bu_t \\ &= Ax_t + Bu_{t-i} \\ &= Ax_t + M_5(BKMC + LC)x_{t-i} \\ &\quad + BK(A + BK)^{i-1}M_4\hat{x}_{t-i|t-i-1} \end{aligned} \quad (19)$$

It follows from (2) and (3) that the state observer has the following form

$$\begin{aligned} \hat{x}_{t+1|t} &= A\hat{x}_{t|t-1} + Bu_{t|t} + L(Cx_t - C\hat{x}_{t|t-1}) \\ &= (A - LC)\hat{x}_{t|t-1} + LCx_t + BK(\hat{x}_{t|t-1} \\ &\quad + M(y_t - C\hat{x}_{t|t-1})) \\ &= M_4\hat{x}_{t|t-1} + (BKMC + LC)x_t \end{aligned} \quad (20)$$

Let $X(t) = \begin{bmatrix} x_t^T & x_{t-1}^T & \cdots & x_{t-i}^T & \hat{x}_{t-i|t-i}^T \end{bmatrix}^T$, then,

the augmented system can be expressed as

$$X(t+1) = \begin{bmatrix} A & 0 & 0 & \cdots & 0 & M_1 & M_2 \\ I & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & I & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & I & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & M_3 & M_4 \end{bmatrix} X(t)$$

Let the state transformation $X'(t) = \begin{bmatrix} x_t'^T & x_{t-1}'^T & \cdots & x_{t-i}'^T & \hat{x}_{t-i|t-i}'^T \end{bmatrix}^T$ and

$$X'(t) = \begin{bmatrix} I & -A & 0 & \cdots & 0 & 0 & 0 \\ 0 & I & -A & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & I & -A & 0 \\ 0 & 0 & 0 & \cdots & 0 & I & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & I \end{bmatrix} X(t)$$

then,

$$X'(t+1) = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & M_1 & M_2 \\ I & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & I & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & I & A & 0 \\ 0 & 0 & 0 & \cdots & 0 & M_3 & M_4 \end{bmatrix} X'(t) \quad (21)$$

Let the state transformation $X''(t) = \begin{bmatrix} x_t''^T & x_{t-1}''^T & \cdots & x_{t-i}''^T & \hat{x}_{t-i|t-i}''^T \end{bmatrix}^T$, and application of the state transformation

$$X''(t) = \begin{bmatrix} I & 0 & 0 & \cdots & 0 & 0 & -M_5 \\ 0 & I & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & I & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & I & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & I \end{bmatrix} X'(t)$$

results in

$$X''(t+1) = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ I & 0 & 0 & \cdots & 0 & 0 & M_5 \\ 0 & I & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & I & A & 0 \\ 0 & 0 & 0 & \cdots & 0 & M_3 & M_4 \end{bmatrix} X''(t) \quad (22)$$

From the structure of equation (22), it can be concluded that the system is stable if and only if the following system is stable

$$Z(t+1) = \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 & M_5 \\ I & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & I & A & 0 \\ 0 & 0 & \cdots & 0 & M_3 & M_4 \end{bmatrix} Z(t) \quad (23)$$

where $Z(t) = [z_1^T(t) \ z_2^T(t) \ \cdots \ z_i^T(t) \ z_{i+1}^T(t)]^T$.
By the state transformation

$$Z'(t) = \begin{bmatrix} I & 0 & \cdots & 0 & 0 & 0 \\ 0 & I & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & I & -I \\ 0 & 0 & \cdots & 0 & 0 & I \end{bmatrix} Z(t)$$

the following system is derived

$$Z'(t+1) = \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 & M_5 \\ I & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & I & M_6 & -BK \\ 0 & 0 & \cdots & 0 & M_3 & A+BK \end{bmatrix} Z'(t) \quad (24)$$

The system matrices of system (23) and system (24) are just in the form of (17) and (18). Therefore, the closed-loop system is stable if and only if all eigenvalues of matrices (17) and (18) are within the unit circle. ■

Corollary 2: For networked predictive control systems (1) with constant network delay $i \in Z^+$, the closed-loop system is stable if and only if one of the following time-delay system is stable.

$$\begin{aligned} x_1(t+1) &= Ax_1(t) + BK(A+BK)^{i-1}x_2(t-i+1) \\ x_2(t+1) &= (BKMC+LC)x_1(t) + M_4x_2(t) \end{aligned} \quad (25)$$

and

$$\begin{aligned} x_1(t+1) &= M_6x_1(t) - BKx_2(t) + M_5x_2(t-i+1) \\ x_2(t+1) &= (BKMC+LC)x_1(t) + (A+BK)x_2(t) \end{aligned} \quad (26)$$

Proof: From the proof of theorem 1, the closed-loop system is stable if and only if system (23) or system (24) is stable. Without loss of generalization, it will be proved that the closed-loop system is stable if and only if system (25) is stable. From system (23) it follows that

$$\begin{aligned} z_1(t+1) &= BK(A+BK)^{i-1}z_{i+1}(t) \\ z_2(t+1) &= BK(A+BK)^{i-1}z_{i+1}(t-1) \\ &\vdots \\ z_{i-1}(t+1) &= BK(A+BK)^{i-1}z_{i+1}(t-i+2) \end{aligned} \quad (27)$$

then

$$\begin{aligned} z_i(t+1) &= Az_i(t) + z_{i-1}(t) \\ &= Az_i(t) + BK(A+BK)^{i-1}z_{i+1}(t-i+1) \end{aligned} \quad (28)$$

combined with equation

$$z_{i+1}(t+1) = (BKMC+LC)z_i(t) + M_4z_{i+1}(t) \quad (29)$$

it follows that equations (28) and (29) are equivalent to system (25). ■

To facilitate the presentation of the main results, the following Lemmas are introduced.

Lemma 1: [14]. For the system

$$x(k+1) = Ax(k) + A_1x(k-i) \quad (30)$$

it is asymptotically stable for any delay i if the following conditions are satisfied.

$$\|A\| + \|A_1\| < 1 \quad (31)$$

where $\|\cdot\|$ stands for the Euclidean norm of a vector and its induced norm of a matrix.

With Lemma 1 stated above, there are the following results.

Corollary 3: i) If K , L and M are designed such that

$$\left\| \begin{bmatrix} A & 0 \\ BKMC+LC & M_4 \end{bmatrix} \right\| + \|M_5\| < 1 \quad (32)$$

then (25) is stable for $\forall i \in Z^+$;

ii) If K , L and M are designed such that

$$\left\| \begin{bmatrix} M_6 & -BK \\ BKMC+LC & A+BK \end{bmatrix} \right\| + \|M_5\| < 1 \quad (33)$$

then (26) is asymptotically stable for $\forall i \in Z^+$.

Proof: System (25) can be written as

$$\begin{bmatrix} x_1(t+1) \\ x_2(t+1) \end{bmatrix} = \Omega \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \Omega_1 \begin{bmatrix} x_1(t-i+1) \\ x_2(t-i+1) \end{bmatrix} \quad (34)$$

where $\Omega = \begin{bmatrix} A & 0 \\ BKMC+LC & M_4 \end{bmatrix}$, $\Omega_1 = \begin{bmatrix} 0 & M_5 \\ 0 & 0 \end{bmatrix}$.

By Lemma 1, system (25) is asymptotically stable if (32) holds. In the same way, system (26) is asymptotically stable under condition (33). ■

Corollary 4: For networked predictive control system (1) with constant network delay $i \in Z^+$, the closed-loop system is stable if and only if the non-zero solutions of the following equation are within the unit circle

$$\begin{vmatrix} z^{i-1}(zI-A) & -M_5 \\ -M_3 & zI-M_4 \end{vmatrix} = 0 \quad (35)$$

Proof: The characteristic equation of system (23) can be described as follows:

$$\det \left(\begin{bmatrix} zI & 0 & \cdots & 0 & 0 & -M_5 \\ -I & zI & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & -I & zI-A & 0 \\ 0 & 0 & \cdots & 0 & -M_3 & zI-M_4 \end{bmatrix} \right) = 0 \quad (36)$$

By the property of the determinant, that is, for matrices $\Pi_{11}, \Pi_{12}, \Pi_{21}, \Pi_{22}$ with appropriate dimensions and $\det(\Pi_{11}) \neq 0$

$$\det \left(\begin{bmatrix} \Pi_{11} & \Pi_{12} \\ \Pi_{21} & \Pi_{22} \end{bmatrix} \right) = \det(\Pi_{11}) \det(\Pi_{22} - \Pi_{21} \Pi_{11}^{-1} \Pi_{12}) \quad (37)$$

Let

$$\Pi_{11} = \begin{bmatrix} zI & 0 & \cdots & 0 \\ -I & zI & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & -I & zI \end{bmatrix}, \quad \Pi_{12} = \begin{bmatrix} 0 & -M_5 \\ 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \end{bmatrix}$$

$$\Pi_{21} = \begin{bmatrix} 0 & \cdots & -I \\ 0 & \cdots & 0 \end{bmatrix}, \Pi_{22} = \begin{bmatrix} zI - A & 0 \\ -M_3 & zI - M_4 \end{bmatrix}$$

Then

$$\Pi_{21}\Pi_{11}^{-1}\Pi_{12} = \begin{bmatrix} 0 & M_5 \\ 0 & 0 \end{bmatrix} \quad (38)$$

So, the left side equation can be written in the following form

$$\begin{aligned} & \text{Det} \left(\begin{bmatrix} zI & 0 & \cdots & 0 & 0 & -M_5 \\ -I & zI & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & -I & zI - A & 0 \\ 0 & 0 & \cdots & 0 & -M_3 & zI - M_4 \end{bmatrix} \right) \\ &= \det(\Pi_{11})\det(\Pi_{22} - \Pi_{21}\Pi_{11}^{-1}\Pi_{12}) \\ &= s^{n(i-1)}\det \left(\begin{bmatrix} zI - A & -M_5 \\ -M_3 & zI - M_4 \end{bmatrix} \right) \end{aligned} \quad (39)$$

On the other hand,

$$\begin{aligned} & \begin{vmatrix} z^{i-1}(zI - A) & -M_5 \\ -M_3 & zI - M_4 \end{vmatrix} \\ &= \begin{vmatrix} z^{i-1}I & 0 \\ 0 & I \end{vmatrix} \begin{vmatrix} zI - A & -M_5 \\ -M_3 & zI - M_4 \end{vmatrix} \\ &= z^{n(i-1)}\det \left(\begin{bmatrix} zI - A & -M_5 \\ -M_3 & zI - M_4 \end{bmatrix} \right) \end{aligned} \quad (40)$$

Therefore, it is clear from (39) and (40) that the conclusion of Corollary 4 is correct. ■

In the following result, the assumption is made that the network delay is random but bounded by N . The network delay compensator is in the form of

$$u_t = u_{t|t-k_t}, \text{ subject to } k_t \leq k_{t-1} + 1 \quad (41)$$

where the network delay k_t is a random number but $k_t \in \{0, 1, 2, \dots, N\}$.

Under this control scheme, the closed-loop system will be a switched linear system, the theory of switched system can be used to design the observer gain L and feedback gain K such that the closed-loop system with random time-delay is stable [15].

In order to analyze the stability of closed-loop system with random time-delay, it is assumed that the random time-delay is k_t at t step, then, it follows from equations (19) and (20) that the augmented system is

$$\bar{x}(t+1) = \bar{A}(k_t)\bar{x}(t) \quad (42)$$

where

$$\begin{aligned} \bar{x}^T(t) &= [x_t^T \ x_{t-1}^T \ \cdots \ x_{t-k_t}^T \ x_{t-(k_t+1)}^T \ \cdots \ x_{t-N}^T \ \hat{x}_{t|t-1}^T \\ &\quad \hat{x}_{t-1|t-2}^T \ \cdots \ \hat{x}_{t-k_t|t-k_t-1}^T \ \cdots \ \hat{x}_{t-N|t-N-1}^T] \\ \bar{A}(k_t) &= \begin{bmatrix} \bar{A}(k_t)_{11} & \bar{A}(k_t)_{12} & \bar{A}(k_t)_{13} \\ \bar{A}(k_t)_{21} & \bar{A}(k_t)_{22} & \bar{A}(k_t)_{23} \\ \bar{A}(k_t)_{31} & \bar{A}(k_t)_{32} & \bar{A}(k_t)_{33} \\ \bar{A}(k_t)_{41} & \bar{A}(k_t)_{42} & \bar{A}(k_t)_{43} \end{bmatrix} \end{aligned} \quad (43)$$

and $k_t \in \{1, 2, \dots, N\}$, $A(k_t) \in R^{2(N+1)n \times 2(N+1)n}$, $M_1(k_t) = BK(A + BK)^{k_t-1}(BKMC + LC)$,

$M_2(k_t) = BK(A + BK)^{k_t-1}(A + BK - LC - BKMC)$, $M_3(k_t) = BKMC + LC$, $M_4(k_t) = A + BK - LC - BKMC$, $\bar{A}(k_t)_{11} = \begin{bmatrix} A & 0 & \cdots & M_1(k_t) & 0 & \cdots & 0 \end{bmatrix}$, $\bar{A}(k_t)_{12} = \begin{bmatrix} 0 \end{bmatrix}$, $\bar{A}(k_t)_{13} = \begin{bmatrix} 0 & \cdots & M_2(k_t) & 0 & \cdots & 0 & 0 \end{bmatrix}$, $\bar{A}(k_t)_{21} = I$, $\bar{A}(k_t)_{22} = 0$, $\bar{A}(k_t)_{23} = 0$, $\bar{A}(k_t)_{31} = \begin{bmatrix} M_3(k_t) & 0 & \cdots & 0 & 0 & \cdots & 0 \end{bmatrix}$, $\bar{A}(k_t)_{32} = 0$, $\bar{A}(k_t)_{33} = \begin{bmatrix} M_4(k_t) & \cdots & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}$, $\bar{A}(k_t)_{41} = 0$, $\bar{A}(k_t)_{42} = 0$, $\bar{A}(k_t)_{43} = \begin{bmatrix} I & 0 \end{bmatrix}$. In the case of the closed-loop system without time-delay (13), it can also be extended as

$$\bar{x}(t+1) = \bar{A}(0)\bar{x}(t) \quad (44)$$

where

$$\bar{A}(0) = \begin{bmatrix} \bar{A}(0)_{11} & \bar{A}(0)_{12} & \bar{A}(0)_{13} \\ \bar{A}(0)_{21} & \bar{A}(0)_{22} & \bar{A}(0)_{23} \\ \bar{A}(0)_{31} & \bar{A}(0)_{32} & \bar{A}(0)_{33} \\ \bar{A}(0)_{41} & \bar{A}(0)_{42} & \bar{A}(0)_{43} \end{bmatrix} \quad (45)$$

and $\bar{A}(0) \in R^{2(N+1)n \times 2(N+1)n}$, $M_1(0) = A + BKMC$, $M_2(0) = BK - BKMC$, $M_3(0) = BKMC + LC$, $M_4(0) = A + BK - LC - BKMC$, $\bar{A}(0)_{11} = \begin{bmatrix} A & 0 & \cdots & M_1(0) & 0 & \cdots & 0 \end{bmatrix}$, $\bar{A}(0)_{12} = \begin{bmatrix} 0 \end{bmatrix}$, $\bar{A}(0)_{13} = \begin{bmatrix} M_2(0) & \cdots & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}$, $\bar{A}(0)_{21} = I$, $\bar{A}(0)_{22} = 0$, $\bar{A}(0)_{23} = 0$, $\bar{A}(0)_{31} = \begin{bmatrix} M_3(0) & 0 & \cdots & 0 & 0 & \cdots & 0 \end{bmatrix}$, $\bar{A}(0)_{32} = 0$, $\bar{A}(0)_{33} = \begin{bmatrix} M_4(0) & \cdots & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}$, $\bar{A}(0)_{41} = 0$, $\bar{A}(0)_{42} = 0$, $\bar{A}(0)_{43} = \begin{bmatrix} I & 0 \end{bmatrix}$.

Thus, the stability of closed-loop system with random time-delay can be stated as follows.

Theorem 5: For the networked predictive control systems with random network delay in the forward channel, the closed-loop system is stable if there exists a positive definite matrix $P \in R^{2(N+1)n \times 2(N+1)n}$ such that

$$\bar{A}^T(k_t)P\bar{A}(k_t) - P < 0 \quad (46)$$

for $k_t = 0, 1, 2, \dots, N$

Proof: From the extended closed-loop system with random time-delay (42) and the extended closed-loop system without time-delay (44), it follows that the closed-loop system is a switched system which is composed of $N+1$ discrete-time subsystems, i.e.,

$$\bar{x}_{t+1} = \bar{A}(k_t)\bar{x}_t \quad (47)$$

where $k_t = 0, 1, \dots, N$. The switched system can be described as

$$\bar{x}_{t+1} = \bar{A}_{\delta(k)}\bar{x}_t \quad (48)$$

where $\delta(k) : \{0, 1, \dots\} \rightarrow \{0, 1, 2, \dots, N\}$, $\delta(k)$ is switching signal.

Let $V_t = \bar{x}_t^T P \bar{x}_t$, then

$$\begin{aligned} V_{t+1} - V_t &= \bar{x}_{t+1}^T P \bar{x}_{t+1} - \bar{x}_t^T P \bar{x}_t \\ &= \bar{x}_t^T [\bar{A}_{\delta(k)}^T P \bar{A}_{\delta(k)} - P] \bar{x}_t \end{aligned} \quad (49)$$

From (46), it follows that $V_{t+1} - V_t < 0$, for $\delta(k)$. Therefore, system (48) is stable for all switching sequences $\delta(k)$. ■

Remark 2: It can be deduced easily from (46) that each subsystem is stable. If K , L and M are designed previously, then (46) is a set of LMIs. LMI toolbox can be used to find a feasible solution P

IV. CONCLUSIONS

A new networked predictive control scheme in a state-space form has been proposed for MIMO networked distributed control systems with random network delay and the stability of the closed-loop networked predictive control systems has also been discussed in this paper. Based on the network feature of transmitting a set of data each time, the proposed networked predictive controller consists of the control prediction generator and the network delay compensator. The former provides a set of future control predictions to satisfy the system performance requirements. The latter compensates the random network transmission delay. Two important theorems on the stability of the closed-loop networked predictive control system have given the analytical stability criteria for the fixed and random network delays, respectively, which have successfully solved the stochastic problem in a deterministic way. The state-space design approach for NCFS can also be extended to the design of networked control systems with network delay in the feedback channel.

ACKNOWLEDGMENT

The authors would like to thank the referees for their valuable and helpful comments which have improved the presentation.

REFERENCES

- [1] J. W. Overstreet and A. Tzes, "An internet-based real-time control engineering laboratory," *IEEE Control Systems Magazine*, vol. 19, pp. 19–34, 1999.
- [2] S. H. Yang, X. chen, D. W. Edwards, and J. L. Alty, "Design issues and implementation of internet based process control," *Control Engineering Practices*, vol. 11(6), pp. 709–720, 2003.
- [3] P. V. Zhivoglyadov and R. H. Middleton, "Networked control design for linear systems," *Automatica*, vol. 39(4), pp. 743–750, 2003.
- [4] W. Zhang, M. S. Brannicky, and S. M. Philips, "Stability of networked control systems," *IEEE Control Systems Magazine*, vol. 21(1), pp. 84–99, 2001.
- [5] L. A. Montestruque and P. J. Antsaklis, "On the model-based control of networked systems," *Automatica*, vol. 39(10), pp. 1837–1843, 2003.
- [6] B. Aktan, C. A. Bohus, L. A. Crowl, and M. H. Shor, "Distance learning applied to control engineering laboratories," *IEEE Transactions on Education*, vol. 39, pp. 320–326, 1996.
- [7] A. Halley and P. Gauld, "Integration where java meets process control," *control and instrumentation*, vol. 31(4), pp. 57–58, 1999.
- [8] S. Wong and R. W. Brockett, "Systems with finite communication bandwidth constraints ii: stabilization with limited information feedback," *IEEE Transaction on Automatic Control*, vol. 44(5), pp. 1049–1053, 1999.
- [9] M. Cushing, "Process control across the internet," *Chemical Engineering*, pp. 80–82, 2000.
- [10] K. Taylor and B. Dalton, "Internet robots: a new robotics niche," *IEEE Robotics and Automation Magazine*, vol. 3, 2000.
- [11] Y. Nemoto, N. Hamamoto, R. Suzuki, T. Ikegami, Y. Hashimoto, T. Ide, K. Ohta, G. Mansfield, and N. Kato, "Construction and utilization experiment of multimedia education system using satellite ets-v and internet," *IEICE Transaction on Information and systems*, vol. 2, pp. 162–169, 1997.
- [12] J. Naylor, "Online data transfer standard," *The Chemical Engineers*, vol. 24, 2002.
- [13] G. P. Liu, J. X. MU, D. Rees, and S. C. Chai, "Design and stability analysis of networked control systems with random communication time delay using the modified mpc," *International Journal of Control*, vol. 79, p. 288C297, 2006.
- [14] T. Mori, N. Fukuma, and M. Kuwahara, "Delay-independent stability criteria for discrete-delay systems," *IEEE Transactions on Automatic Control*, vol. 27, no. 4, 1982.
- [15] R. A. Decarlo, M. S. Branicky, S. Pettersson, and B. Lennartson, "Perspectives and results on the stability and stabilizability of hybrid systems," *Proceedings of the IEEE*, vol. 88, pp. 1069–1082, 2000.